INVERTIS UNIVERSITY, BAREILLY

B.Tech CSE – Third Year | DBMS & SQL Lab (BCS552)

EXPERIMENT NO. 7

1. Experiment Title:

Using Aggregate Functions

2. Aim:

Use aggregate functions to count sales rows, calculate a total, and find the highest amount.

3. Software / Tools Required:

SQL database system (such as MySQL, PostgreSQL, or SQLite where supported), SQL editor/IDE, and a computer.

4. Theory:

Aggregate functions calculate a single summary value from multiple rows. COUNT counts rows or non-NULL values, SUM adds numeric values, and MAX returns the largest value.

5. Algorithm / Procedure:

1. Create the sales table and insert the four supplied sales records.
2. Use COUNT(*) and alias the result as num_sales.
3. Use SUM(amount) and alias the result as total.
4. Use MAX(amount) and alias the result as highest.

6. Program / Queries / Commands:

```
CREATE TABLE sales (
    salesperson VARCHAR(20),
    region VARCHAR(20),
    amount INT
);

INSERT INTO sales (salesperson, region, amount)
VALUES
    ('Deepak', 'North', 12000),
    ('Pooja', 'South', 8500),
    ('Deepak', 'South', 9500),
    ('Lata', 'North', 11000);

-- Task 1
SELECT COUNT(*) AS num_sales FROM sales;

-- Task 2
SELECT SUM(amount) AS total FROM sales;

-- Task 3
SELECT MAX(amount) AS highest FROM sales;
```

7. Input:

Amounts: 12000, 8500, 9500, 11000.

8. Output:

Expected output (column spacing and display format may differ in your SQL tool):

Task 1: num_sales
4
Task 2: total
41000

Task 3: highest
12000

9. Result:

The SQL statements for Experiment 7 were prepared to perform the stated tasks. On executing the setup and queries successfully, the expected results shown above should be obtained.

10. Viva-Voce Questions:

1. What is an aggregate function?
2. What does COUNT(*) return?
3. What does SUM() calculate?
4. What does MAX() return?